Lecture 20

Type A branching and Gelfand–Tsetlin patterns

Objectives. *We restrict the polynomial $\mathfrak{gl}_n$-modules of Lecture 18 to the upper-left block $\mathfrak{gl}_{n-1}$. Each summand occurs at most once. Repeating this along a chain of blocks splits a module into canonical lines. Restriction to $\mathfrak{sl}_{n-1}$ alone can repeat a summand.*

By the end of this lecture you will be able to:

- List the summands of $\mathbb{S}_\lambda(\mathbb{C}^n)$ restricted to $\mathfrak{gl}_{n-1}$ by interlacing.
- Prove this branching rule from the Schur interlacing identity and central scalars.
- Construct Gelfand–Tsetlin lines and state how far a basis is determined.
- Explain the two copies of V_1 in the restriction of the $\mathfrak{sl}_3$ adjoint module to $\mathfrak{sl}_2$.

Throughout, $n \geq 2$, all modules are finite-dimensional over $\mathbb{C}$, and E_{ij} is a matrix unit. For $1 \leq k \leq n$, let

$$\mathfrak{gl}_k = \operatorname{span}\{E_{ij} : 1 \leq i, j \leq k\} \subseteq \mathfrak{gl}_n, \qquad Z_k = E_{11} + \cdots + E_{kk}.$$

Here $\mathfrak{gl}_k$ is the upper-left $k \times k$ block. Its centre is $\mathbb{C}Z_k$, and

$$\mathfrak{gl}_k = \mathfrak{sl}_k \oplus \mathbb{C}Z_k, \qquad \mathfrak{sl}_1 = 0.$$

Let λ be a partition with at most n parts and size $d = |\lambda|$. Use the image model of Lecture 18 to realise the polynomial module $\mathbb{S}_\lambda(\mathbb{C}^n)$ as a submodule of $(\mathbb{C}^n)^{\otimes d}$. A Schur module inside a tensor power is polynomial in the sense of Lecture 18. § 20.5 also treats one twisted module that is not polynomial.

For a $\mathfrak{gl}_n$-module V and $a \in \mathbb{Z}^n$, the weight space is

$$V_a = \{v \in V : E_{ii}v = a_i v \text{ for } 1 \leq i \leq n\}.$$

If V is the sum of its weight spaces, its character is

$$\operatorname{ch} V = \sum_a \dim(V_a)\, x_1^{a_1} \cdots x_n^{a_n}.$$

By Lecture 18, $\operatorname{ch} \mathbb{S}_\lambda(\mathbb{C}^n) = s_\lambda(x_1, \ldots, x_n)$, the Schur polynomial.

20.1 The interlacing identity

Definition 20.1 (Interlacing). Let $\lambda = (\lambda_1, \ldots, \lambda_n)$ be a partition, padded by zeros. An

integer sequence $\mu = (\mu_1, \dots, \mu_{n-1})$ *interlaces* λ, written $\mu \prec \lambda$, if

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots \geq \lambda_{n-1} \geq \mu_{n-1} \geq \lambda_n.$$

Such a μ is a partition with at most $n-1$ parts, since $\mu_{n-1} \geq \lambda_n \geq 0$.

Theorem 20.2 (Interlacing identity · *C*)**.** *For a partition λ with at most n parts,*

$$s_\lambda(x_1, \dots, x_n) = \sum_{\mu \prec \lambda} s_\mu(x_1, \dots, x_{n-1})\, x_n^{|\lambda|-|\mu|}. \tag{20.1}$$

Source: the finite-group course, Lecture 14 and its optional sheet. The tableau formula used below also appears in William Fulton and Joe Harris, Representation Theory: A First Course*, Exercise A.31(a).*

The tableau formula from the same lecture gives a direct derivation [C]:

$$s_\lambda(x_1, \dots, x_n) = \sum_T x^T.$$

The sum runs over semistandard tableaux T of shape λ with entries in $\{1, \dots, n\}$. The exponent of x_i in x^T is the number of entries i.

Delete the entries n from T. Rows weakly increase, so these entries end their rows. Columns strictly increase, so each column has at most one of them, at its bottom. What remains is a semistandard tableau T' of some shape $\mu \subseteq \lambda$ with entries at most $n-1$. Hence μ has at most $n-1$ rows.

The removed boxes lie in distinct columns. This holds exactly when

$$\mu_i \geq \lambda_{i+1} \quad \text{for } 1 \leq i \leq n-1.$$

If $\mu_i < \lambda_{i+1}$, column $\mu_i + 1$ loses boxes in rows i and $i+1$. If rows $i < i'$ both lose a box in column j, then $\mu_i < j \leq \lambda_{i'} \leq \lambda_{i+1}$. Together with $\mu \subseteq \lambda$, the condition is $\mu \prec \lambda$.

Conversely, filling λ/μ with n turns any such T' into a valid T. The monomials satisfy

$$x^T = x^{T'}\, x_n^{|\lambda|-|\mu|}.$$

Summing over $\mu \prec \lambda$ and over T' of shape μ, and using the tableau formula in $n-1$ variables, gives (20.1).

Example 20.3 (Interlacing for $(2,1,0)$)**.** For $\lambda = (2,1,0)$ the conditions are $2 \geq \mu_1 \geq 1 \geq \mu_2 \geq 0$. Thus

$$\mu \in \{(2,1),\ (2,0),\ (1,1),\ (1,0)\},$$

and

$$s_{(2,1)}(x_1,x_2,x_3) = s_{(2,1)}(x_1,x_2) + \big(s_{(2)}(x_1,x_2) + s_{(1,1)}(x_1,x_2)\big)x_3 + s_{(1)}(x_1,x_2)\,x_3^2.$$

At $x_1 = x_2 = x_3 = 1$ this reads $8 = 2 + (3+1) + 2$.

20.2 Branching from $\mathfrak{gl}_n$ to $\mathfrak{gl}_{n-1}$

Lemma 20.4 (Weights and central scalars)**.** *Let $m \geq 1$, let λ have at most m parts, and let $V = \mathbb{S}_\lambda(\mathbb{C}^m)$.*

(a) V is the sum of its weight spaces. Every weight a has $a_i \in \mathbb{Z}_{\geq 0}$ and $a_1 + \cdots + a_m = |\lambda|$.
(b) The central element $Z_m = I_m$ acts on V by the scalar $|\lambda|$.

Proof. For a standard basis tensor,

$$E_{jj}\,(e_{i_1} \otimes \cdots \otimes e_{i_d}) = \#\{r : i_r = j\}\; e_{i_1} \otimes \cdots \otimes e_{i_d}.$$

So the diagonal matrices act on $(\mathbb{C}^m)^{\otimes d}$ by commuting diagonalisable operators, with weights in $\mathbb{Z}^m_{\geq 0}$ of total $d = |\lambda|$. Commuting diagonalisable operators stay simultaneously diagonalisable on an invariant subspace, and the submodule V is invariant under them. This proves (a). Summing the entries of a weight gives (b). For $v \in V_a$,

$$Z_m v = (E_{11} + \cdots + E_{mm})v = (a_1 + \cdots + a_m)v = |\lambda| v.$$ ∎

Lemma 20.5 (Independence of Schur polynomials)**.** *Let $m \geq 1$. The polynomials $s_\mu(x_1, \ldots, x_m)$, for partitions μ with at most m parts, are linearly independent.*

Proof. For $m \geq 2$, this is the independence lemma of Lecture 19, proved by comparing the strictly decreasing monomials in the alternants. For $m = 1$, the Schur polynomials are $s_{(j)}(x_1) = x_1^j$ for integers $j \geq 0$. Distinct monomials are linearly independent. ∎

Theorem 20.6 (Branching from $\mathfrak{gl}_n$ to $\mathfrak{gl}_{n-1} \cdot P$ given the interlacing identity)**.** *Let λ have at most n parts and $d = |\lambda|$. As a $\mathfrak{gl}_{n-1}$-module,*

$$\mathbb{S}_\lambda(\mathbb{C}^n) \cong \bigoplus_{\mu \prec \lambda} \mathbb{S}_\mu(\mathbb{C}^{n-1}),$$

each summand occurring once. The summand for μ is the $\mathfrak{gl}_{n-1}$-isotypic component of type μ, and E_{nn} acts on it by $d - |\mu|$.

Proof. Write $V = \mathbb{S}_\lambda(\mathbb{C}^n)$ and $Z = Z_{n-1}$.

Step 1: split by the centre of $\mathfrak{gl}_{n-1}$. By Lemma 20.4(a), Z is diagonalisable on V with eigenvalues in $\mathbb{Z}_{\geq 0}$:

$$V = \bigoplus_{k \geq 0} V[k], \qquad V[k] = \{v \in V : Zv = kv\}.$$

Since Z commutes with $\mathfrak{gl}_{n-1}$, each $V[k]$ is a $\mathfrak{gl}_{n-1}$-submodule. Lemma 20.4(b) gives

$$E_{nn} = I_n - Z = d - k \quad \text{on } V[k].$$

Step 2: irreducible pieces. By Weyl's theorem (Lecture 5), $V[k]$ is a direct sum of irreducible $\mathfrak{sl}_{n-1}$-modules U. Since Z acts on U by k, each U is an irreducible $\mathfrak{gl}_{n-1}$-submodule. For $n = 2$, take the U to be lines.

Step 3: identify each piece. The highest weight space of U is one-dimensional (Lecture 12). The diagonal of $\mathfrak{gl}_{n-1}$ is spanned by the diagonal of $\mathfrak{sl}_{n-1}$ and Z, so it preserves this space. Hence a highest weight vector u satisfies

$$E_{ii}u = \mu_i u \quad (1 \leq i \leq n-1), \qquad E_{nn}u = (d-k)u.$$

So u is a weight vector of V of weight $(\mu, d-k)$. Lemma 20.4(a), applied to this weight, gives

$$\mu_i \in \mathbb{Z}_{\geq 0}, \qquad \mu_1 + \cdots + \mu_{n-1} = d - (d-k) = k.$$

Dominance (Lecture 12) gives

$$\mu_i - \mu_{i+1} \in \mathbb{Z}_{\geq 0}, \qquad 1 \leq i \leq n-2.$$

So μ is a partition of k with at most $n-1$ parts. By Lecture 18, $\mathbb{S}_\mu(\mathbb{C}^{n-1})$ is an irreducible $\mathfrak{sl}_{n-1}$-module with the same highest weight, and Z acts on it by $|\mu| = k$. By Lecture 13 there is an $\mathfrak{sl}_{n-1}$-isomorphism $U \cong \mathbb{S}_\mu(\mathbb{C}^{n-1})$. It commutes with Z, which is the scalar k on both sides, so it is a $\mathfrak{gl}_{n-1}$-isomorphism.

Step 4: count by characters. Steps 1–3 give multiplicities $m_\mu \geq 0$ with

$$V \cong \bigoplus_\mu \mathbb{S}_\mu(\mathbb{C}^{n-1})^{\oplus m_\mu},$$

and E_{nn} acts by $d - |\mu|$ on the copies of type μ. A vector of $\mathfrak{gl}_{n-1}$-weight b in such a copy has $\mathfrak{gl}_n$-weight $(b, d-|\mu|)$. Therefore

$$s_\lambda(x_1, \dots, x_n) = \sum_\mu m_\mu \, s_\mu(x_1, \dots, x_{n-1}) \, x_n^{d-|\mu|}.$$

Subtract (20.1). Put $\delta_\mu = 1$ if $\mu \prec \lambda$ and $\delta_\mu = 0$ otherwise. The coefficient of x_n^{d-k} gives

$$\sum_{|\mu|=k} (m_\mu - \delta_\mu) \, s_\mu(x_1, \dots, x_{n-1}) = 0 \quad \text{for each } k.$$

Lemma 20.5 gives $m_\mu = \delta_\mu$.

Step 5: isotypic components. Distinct μ give distinct $\mathfrak{gl}_{n-1}$-highest weights, so the summands are pairwise nonisomorphic. A type occurring once has that summand as its isotypic component (Lecture 14). ∎

20.3 Gelfand–Tsetlin patterns and lines

Apply the branching theorem along the chain

$$\mathfrak{gl}_1 \subset \mathfrak{gl}_2 \subset \cdots \subset \mathfrak{gl}_{n-1} \subset \mathfrak{gl}_n.$$

Definition 20.7 (Gelfand–Tsetlin pattern)**.** A *Gelfand–Tsetlin pattern* with top row λ is a sequence $\Lambda = (\lambda^{(n)}, \dots, \lambda^{(1)})$ with $\lambda^{(n)} = \lambda$ and

$$\lambda^{(k)} \in \mathbb{Z}^k \qquad (1 \leq k \leq n),$$
$$\lambda^{(k-1)} \prec \lambda^{(k)} \qquad (2 \leq k \leq n).$$

Each row $\lambda^{(k)}$ is a partition, by descending induction on k and the remark after the definition of interlacing. For $n = 3$ a pattern is written as a triangle, each entry between its upper neighbours:

$$\begin{array}{ccccc} \lambda_1 & & \lambda_2 & & \lambda_3 \\ & \lambda_1^{(2)} & & \lambda_2^{(2)} & \\ & & \lambda_1^{(1)} & & \end{array}$$

Fix $V = \mathbb{S}_\lambda(\mathbb{C}^n)$ and a pattern Λ. Define $V_\Lambda^{(n)} = V$. For $k < n$, let $V_\Lambda^{(k)}$ be the $\mathfrak{gl}_k$-isotypic component of type $\lambda^{(k)}$ in $V_\Lambda^{(k+1)}$. It depends only on rows $n, \dots, k$. Put $L_\Lambda = V_\Lambda^{(1)}$. A module isomorphism preserves every isotypic component, so it preserves these lines.

Theorem 20.8 (Gelfand–Tsetlin decomposition · *P*). *(a)* $V_\Lambda^{(k)} \cong \mathbb{S}_{\lambda^{(k)}}(\mathbb{C}^k)$ *as* $\mathfrak{gl}_k$*-modules.*

(b) Each L_Λ *is a line, and* $V = \bigoplus_\Lambda L_\Lambda$ *over all patterns with top row* λ*.*

(c) E_{kk} *acts on* L_Λ *by* $|\lambda^{(k)}| - |\lambda^{(k-1)}|$*, where* $|\lambda^{(0)}| = 0$*.*

In particular, $\dim \mathbb{S}_\lambda(\mathbb{C}^n)$ *is the number of patterns with top row* λ*.*

Proof. (a) Use descending induction on k. The case $k = n$ is the definition. Suppose $V_\Lambda^{(k+1)} \cong \mathbb{S}_{\lambda^{(k+1)}}(\mathbb{C}^{k+1})$. A module isomorphism carries $\mathfrak{gl}_k$-isotypic components to $\mathfrak{gl}_k$-isotypic components. Theorem 20.6, with n replaced by $k+1$, therefore gives

$$V_\Lambda^{(k+1)} = \bigoplus_{\kappa \prec \lambda^{(k+1)}} V_\Lambda^{(k+1)}[\kappa], \qquad V_\Lambda^{(k+1)}[\kappa] \cong \mathbb{S}_\kappa(\mathbb{C}^k).$$

The component with $\kappa = \lambda^{(k)}$ is $V_\Lambda^{(k)}$.

(b) By (a) with $k = 1$, $L_\Lambda \cong \mathbb{S}_{\lambda^{(1)}}(\mathbb{C}^1)$, which is one-dimensional. The display in (a) splits $V_\Lambda^{(k+1)}$ over the possible rows k. Iterating from $k = n-1$ down to $k = 1$ gives $V = \bigoplus_\Lambda L_\Lambda$.

(c) The line L_Λ lies in $V_\Lambda^{(k)}$, on which Z_k acts by $|\lambda^{(k)}|$ by part (a) and Lemma 20.4(b). The same holds for $k-1$ when $k \geq 2$. For $k = 1$ put $Z_0 = 0$. Hence, for $v \in L_\Lambda$,

$$E_{kk}v = (Z_k - Z_{k-1})v = \bigl(|\lambda^{(k)}| - |\lambda^{(k-1)}|\bigr)v. \qquad \blacksquare$$

A *Gelfand–Tsetlin basis* is a choice of nonzero $\xi_\Lambda \in L_\Lambda$ for every pattern. The lines depend only on the chain of blocks, so they are canonical. The vectors are not canonical. Any other choice is

$$\xi'_\Lambda = c_\Lambda \xi_\Lambda, \qquad c_\Lambda \in \mathbb{C}^\times.$$

Write $\rho_V(x)$ for the operator by which $x \in \mathfrak{gl}_n$ acts on V, and $D(x)$ for its matrix in the basis (ξ_Λ). The subscript distinguishes ρ_V from the Weyl vector ρ. Then

$$\rho_V(x)\xi'_{\Lambda'} = c_{\Lambda'} \sum_\Lambda D(x)_{\Lambda\Lambda'}\xi_\Lambda = \sum_\Lambda \frac{c_{\Lambda'}}{c_\Lambda} D(x)_{\Lambda\Lambda'}\,\xi'_\Lambda.$$

So the diagonal entries of $D(x)$ do not change. By part (c) of Theorem 20.8, $D(E_{kk})$ is diagonal in every Gelfand–Tsetlin basis, with the same diagonal entries. Off-diagonal entries of $D(E_{k,k+1})$ depend on the normalisation. Exercise 20.5, an extension of this lecture, proves that $E_{k,k+1}$ changes only row k of a pattern. Exercise 20.6 computes one such operator on the eight lines of the $\mathfrak{sl}_3$ adjoint module.

Distinct patterns may have the same row sums, hence the same weight. So the lines refine the weight spaces, as § 20.5 shows at the zero weight of $\mathfrak{sl}_3$.

20.4 Restriction to $\mathfrak{sl}_{n-1}$

Restriction to $\mathfrak{sl}_{n-1}$ forgets the action of Z_{n-1}.

Proposition 20.9 (Coincidences over $\mathfrak{sl}_{n-1}$ · *P*). *(a) Let μ, μ' be partitions with at most $n-1$ parts. Then $\mathbb{S}_\mu(\mathbb{C}^{n-1}) \cong \mathbb{S}_{\mu'}(\mathbb{C}^{n-1})$ as $\mathfrak{sl}_{n-1}$-modules if and only if $\mu' - \mu = c(1,\dots,1)$ for some $c \in \mathbb{Z}$.*

(b) Let λ have at most n parts. The restriction of $\mathbb{S}_\lambda(\mathbb{C}^n)$ to $\mathfrak{sl}_{n-1}$ is multiplicity-free if and only if $\lambda_i = \lambda_{i+1}$ for some $1 \le i \le n-1$.

Proof. (a) If $n = 2$, then $\mathfrak{sl}_1 = 0$. Both modules are one-dimensional, and every difference is a multiple of (1). Let $n \ge 3$. By Lecture 13, irreducible modules are isomorphic exactly when their highest weights agree. By Lecture 18, the Dynkin labels of $\mathbb{S}_\mu(\mathbb{C}^{n-1})$ are $\mu_i - \mu_{i+1}$. The labels agree exactly when

$$\mu_i - \mu_{i+1} = \mu'_i - \mu'_{i+1} \quad (1 \le i \le n-2),$$

that is, when $\mu'_i - \mu_i$ does not depend on i.

(b) By Theorem 20.6 and (a), the multiplicity of a class is the number of $\mu \prec \lambda$ in one coset $\mu + \mathbb{Z}(1,\dots,1)$. Suppose μ and $\mu + c(1,\dots,1)$ both interlace λ, with $c \ge 1$. For every $i \le n-1$,

$$\lambda_{i+1} \le \mu_i < \mu_i + c \le \lambda_i.$$

So a repeated class forces $\lambda_i > \lambda_{i+1}$ for all $i \le n-1$. Conversely, if all these inequalities hold, then

$$(\lambda_2,\dots,\lambda_n) \quad \text{and} \quad (\lambda_2+1,\dots,\lambda_n+1)$$

both interlace λ and differ by $(1,\dots,1)$. ∎

Over $\mathfrak{gl}_{n-1}$ each isotypic component is irreducible, hence a canonical subspace. Over $\mathfrak{sl}_{n-1}$ a repeated class has a multiplicity space of dimension at least two. Its isotypic component splits into irreducibles in infinitely many ways, and the action of $\mathfrak{sl}_{n-1}$ distinguishes none of them. The operator Z_{n-1} commutes with $\mathfrak{sl}_{n-1}$ and selects one splitting, since it acts by different scalars on the members of one class:

$$|\mu + c(1,\dots,1)| = |\mu| + c(n-1).$$

Inside $\mathfrak{sl}_n$, the traceless element

$$t_n = Z_{n-1} - (n-1)E_{nn}$$

commutes with $\mathfrak{sl}_{n-1}$. By Theorem 20.6 it acts on the μ-summand by

$$|\mu| - (n-1)(d - |\mu|) = n|\mu| - (n-1)d.$$

So t_n separates members of one class. Hence the $\mathfrak{sl}_n$-module $\mathbb{S}_\lambda(\mathbb{C}^n)|_{\mathfrak{sl}_n}$ restricts multiplicity-freely to $\mathfrak{sl}_{n-1} \oplus \mathbb{C}t_n \cong \mathfrak{gl}_{n-1}$.

20.5 The $\mathfrak{sl}_3$ adjoint module restricted to $\mathfrak{sl}_2$

Take $n = 3$ and the standard $\mathfrak{sl}_2$ in the upper-left block:

$$h = E_{11} - E_{22}, \qquad e = E_{12}, \qquad f = E_{21}.$$

By Lecture 18, the adjoint $\mathfrak{sl}_3$-module is the restriction of $\mathbb{S}_{(2,1)}(\mathbb{C}^3)$.

Step 1: $\mathfrak{gl}_2$ labels. Example 20.3 and Theorem 20.6 give

$$\mathbb{S}_{(2,1)}(\mathbb{C}^3) \cong \mathbb{S}_{(2,1)}(\mathbb{C}^2) \oplus \mathbb{S}_{(2)}(\mathbb{C}^2) \oplus \mathbb{S}_{(1,1)}(\mathbb{C}^2) \oplus \mathbb{S}_{(1)}(\mathbb{C}^2),$$

in the order $\mu = (2,1), (2,0), (1,1), (1,0)$. Here

$$\begin{aligned} \dim : &\quad 8 = 2 + 3 + 1 + 2, \\ E_{33} = 3 - |\mu| : &\quad 0,\ 1,\ 1,\ 2. \end{aligned}$$

The four labels μ differ, so this splitting is canonical.

Step 2: the adjoint action of $\mathfrak{gl}_3$. Let $\mathfrak{gl}_3$ act on $\mathfrak{sl}_3$ by ad. The action is nontrivial, but I_3 acts by 0, so this module is not polynomial. Let $\mathbb{C}_{-\operatorname{tr}}$ be the line on which x acts by $-\operatorname{tr} x$. Then

$$\mathfrak{sl}_3 \cong \mathbb{S}_{(2,1)}(\mathbb{C}^3) \otimes \mathbb{C}_{-\operatorname{tr}} \quad \text{as } \mathfrak{gl}_3\text{-modules}.$$

Both sides restrict to the same irreducible $\mathfrak{sl}_3$-module. The identity acts on both by zero:

$$\operatorname{ad}(I_3) = 0, \qquad |(2,1)| - \operatorname{tr} I_3 = 3 - 3 = 0.$$

Hence an $\mathfrak{sl}_3$-isomorphism (Lecture 13) commutes with $\mathfrak{gl}_3 = \mathfrak{sl}_3 \oplus \mathbb{C}I_3$. On $\mathfrak{gl}_2$, the twist shifts every label by $(-1,-1)$ and every E_{33}-eigenvalue by -1. The twist changes no subspace, so it keeps the isotypic components and the Gelfand–Tsetlin lines.

Step 3: explicit summands. Put $t = E_{11} + E_{22} - 2E_{33}$. For $A \in \mathfrak{gl}_2$ and $i \le 2$, matrix multiplication gives

$$[A, E_{i3}] = \sum_{j=1}^{2} A_{ji} E_{j3}, \qquad [A, E_{3i}] = -\sum_{j=1}^{2} A_{ij} E_{3j}.$$

So $\operatorname{span}\{E_{13}, E_{23}\}$ is the $\mathfrak{gl}_2$-module $\mathbb{C}^2$, with highest weight vector E_{13} of weight $(1,0)$. Also $\operatorname{span}\{E_{31}, E_{32}\}$ is its dual, with highest weight vector E_{32} of weight $(0,-1)$. The block $\mathfrak{sl}_2$ is an ideal of $\mathfrak{gl}_2$, with highest weight vector E_{12} of weight $(1,-1)$. Finally, t commutes with $\mathfrak{gl}_2$. These four spans have distinct types and dimensions $2 + 3 + 1 + 2 = 8$, so they are the isotypic components of Step 1. In the table, μ labels the polynomial module of Step 1. The twist of Step 2 gives the $\mathfrak{gl}_2$ label and the eigenvalue of $\operatorname{ad} E_{33}$.

μ	$\mathfrak{gl}_2$ label $\mu - (1,1)$	$\operatorname{ad} E_{33} = 2 - \lvert\mu\rvert$	Gelfand–Tsetlin basis	$\mathfrak{sl}_2$
$(2,1)$	$(1,0)$	-1	$E_{13},\ E_{23}$	V_1
$(2,0)$	$(1,-1)$	0	$E_{12},\ h,\ E_{21}$	V_2
$(1,1)$	$(0,0)$	0	t	V_0
$(1,0)$	$(0,-1)$	1	$E_{32},\ E_{31}$	V_1

Figure 20.1 draws these data in two panels. The upper panel shows each Gelfand–Tsetlin pattern as a path in the branching tree. The path ends at the bottom entry ν, which Step 5 treats. The lower panel places each line at its $\mathfrak{sl}_3$ weight. Its rows are the $\mathfrak{sl}_2$-modules of Step 4.

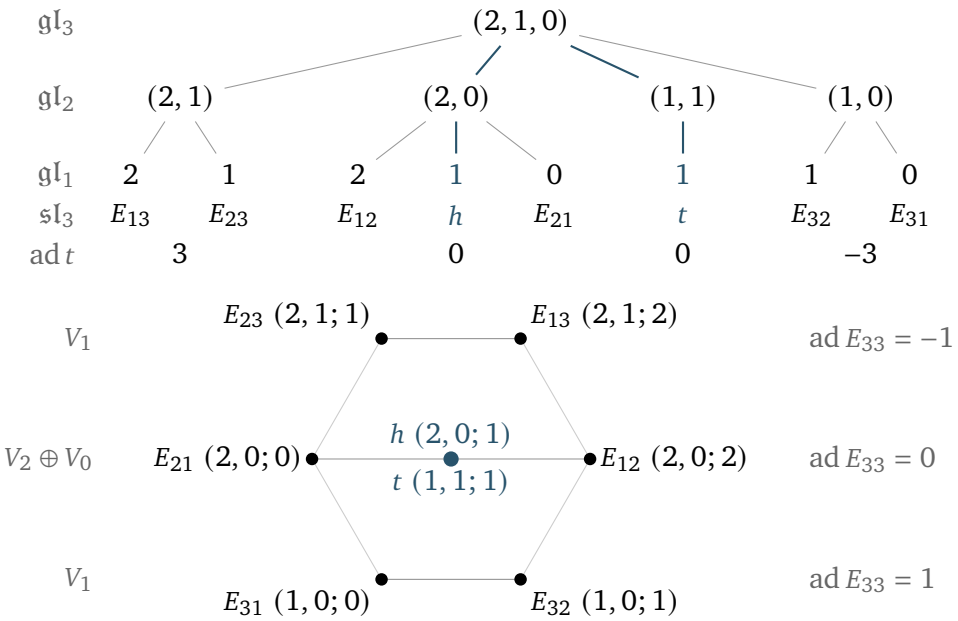


Figure 20.1: Upper panel: paths from $(2,1,0)$ through μ to ν, with the basis vector of each line and the eigenvalue of $\operatorname{ad} t$. Lower panel: the lines at their $\mathfrak{sl}_3$ weights, labelled $(\mu_1,\mu_2;\nu)$, in rows of constant $\operatorname{ad} E_{33}$. The two coloured paths end at weight 0.

Step 4: forget the centre. The $\mathfrak{sl}_2$-label is $\mu_1-\mu_2$, so

$$\mathfrak{sl}_3 \cong V_2 \oplus V_1 \oplus V_1 \oplus V_0, \qquad 8 = 3+2+2+1.$$

The two copies of V_1 have $\mathfrak{gl}_2$ labels $(1,0)$ and $(0,-1)$, which differ by $(1,1)$. Over $\mathfrak{sl}_2$ they are isomorphic. Indeed,

$$\operatorname{ad}(f)E_{13} = E_{23}, \qquad \operatorname{ad}(f)E_{32} = -E_{31},$$
$$\operatorname{ad}(e)E_{23} = E_{13}, \quad \operatorname{ad}(e)(-E_{31}) = E_{32},$$

and $\operatorname{ad}(e)$ maps E_{13} and E_{32} to 0. Define the linear map $T\colon\ \operatorname{span}\{E_{13},E_{23}\} \to \operatorname{span}\{E_{31},E_{32}\}$ by

$$T(E_{13}) = E_{32}, \qquad T(E_{23}) = -E_{31}.$$

It matches highest weight vectors and their f-images in two copies of V_1, so it is an $\mathfrak{sl}_2$-isomorphism. It is not a $\mathfrak{gl}_2$-isomorphism, because Z_2 acts by different scalars:

$$\operatorname{ad}(Z_2)E_{13} = E_{13}, \qquad \operatorname{ad}(Z_2)E_{32} = -E_{32}.$$

The vectors of h-weight 1 that $\operatorname{ad}(e)$ maps to 0 form $\operatorname{span}\{E_{13},E_{32}\}$. Every nonzero vector there generates a copy of V_1. For example, $E_{13}+E_{32}$ generates $\operatorname{span}\{E_{13}+E_{32},\ E_{23}-E_{31}\}$. Inside $\mathfrak{sl}_3$, $t = t_3$ separates the copies:

$$\operatorname{ad}(t)E_{13} = 3E_{13}, \qquad \operatorname{ad}(t)E_{32} = -3E_{32},$$

in agreement with $3|\mu|-6$ for $|\mu|=3$ and $|\mu|=1$. Since $2>1>0$, Proposition 20.9 predicts this repetition.

Step 5: Gelfand–Tsetlin lines. The bottom entry ν satisfies $\mu_1 \ge \nu \ge \mu_2$, giving 2+3+1+2 = 8 patterns. In each block the E_{11}-eigenvalues are $\nu-1$, hence distinct. So the lines are spanned by the table entries, listed with ν decreasing.

By part (c) of Theorem 20.8, the line of the pattern with middle row μ and bottom entry ν has polynomial weight $(\nu,\ |\mu|-\nu,\ 3-|\mu|)$. The twist of Step 2 subtracts $(1,1,1)$. These are the positions in the lower panel of Figure 20.1.

The zero weight space has the two patterns with $|\mu|=2$ and $\nu=1$. They are the two coloured paths of Figure 20.1. These paths differ and have the same final weight, so they give two different canonical lines. The zero weight space splits into these two lines:

$$\mathbb{C}h \oplus \mathbb{C}t.$$

The second line is $\mathbb{C}t$, not $\mathbb{C}(E_{22}-E_{33})$. The matrix entries still depend on the scaling. For example,

$$\operatorname{ad}(e)E_{21} = h = \frac{1}{2}(2h).$$

After replacing h by $2h$, this matrix entry becomes $\frac{1}{2}$.

Self-checks

(1) How many Gelfand–Tsetlin patterns have top row $(1,1,0)$?

(2) Restrict $\mathbb{S}_{(2,2)}(\mathbb{C}^3)$ to $\mathfrak{sl}_2$. Is the result multiplicity-free?

(3) After $\xi_\Lambda \mapsto c_\Lambda \xi_\Lambda$, which product of entries of $D(E_{12})$ and $D(E_{21})$ is unchanged?

Answers. (1) Three: $\mu=(1,1)$ with $\nu=1$, and $\mu=(1,0)$ with $\nu\in\{0,1\}$. This is $\dim \Lambda^2\mathbb{C}^3$. (2) $\mu\in\{(2,2),(2,1),(2,0)\}$ gives $V_0\oplus V_1\oplus V_2$, with $6=1+2+3$. It is multiplicity-free, as $\lambda_1=\lambda_2$. (3) $D(E_{12})_{\Lambda\Lambda'}D(E_{21})_{\Lambda'\Lambda}$, since the factors $c_{\Lambda'}/c_\Lambda$ and $c_\Lambda/c_{\Lambda'}$ cancel.

Summary

Restricting $\mathbb{S}_\lambda(\mathbb{C}^n)$ to $\mathfrak{gl}_{n-1}$ gives each $\mathbb{S}_\mu(\mathbb{C}^{n-1})$ with $\mu\prec\lambda$ exactly once. The proof combines the interlacing identity, complete reducibility and the central scalars $|\mu|$.

Iterating along $\mathfrak{gl}_1\subset\cdots\subset\mathfrak{gl}_n$ gives canonical lines indexed by Gelfand–Tsetlin patterns. Basis vectors in them are fixed only up to independent scalars. Over $\mathfrak{sl}_{n-1}$, summands whose labels differ by full columns become isomorphic, as the two copies of V_1 in $\mathfrak{sl}_3$ show.

Reading

The finite-group course, Lecture 14 and its optional sheet, for the interlacing identity and the tableau formula. William Fulton and Joe Harris, *Representation Theory: A First Course*, §15. Roe Goodman and Nolan R. Wallach, *Symmetry, Representations, and Invariants*, Chapter 8, especially §8.1 on branching for general linear groups. Exercise Sheet 20 gives further calculations.